



CSI473 Project Design Report

Tsa Temo Thuo - Team 10

Phase 1 Interaction, Lifecycle and Integrated Design Report

Field	Complete
Assignment/phase	Phase 1 - Interaction, lifecycle and integrated draft (Lab 5)
Approved project title	A Farm Produce Ordering and Delivery Coordination System for Botswana's Horticulture Sector
Approval decision/date	Approved project baseline - 7 August 2026
Team number	10
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Repository URL	github.com/Shawnjembu/Tsa-Temo-Thuo
Submission tag	phase 1-submission
Date	10 September 2026

1. Executive summary

Tsa Temo Thuo is a farm produce ordering and delivery coordination system designed for Botswana's horticulture sector. The approved problem is a coordination gap between locally available produce and reliable access to buyers: farmers may depend on informal sales channels, smaller buyers may have limited visibility of supply, location and price, and separately arranged transport can introduce delay and coordination risk for perishable produce. The approved project therefore targets improved market access and reduced avoidable post-harvest losses through a digitally coordinated workflow for verified farmers, buyers and transport providers.

This Phase 1 report consolidates the approved requirements baseline with the integrated interaction, lifecycle and domain models. The report is centred on UC-01 - Place Order, the selected vertical-slice use case. Its principal design decision is that stock reservation is an atomic data-layer operation rather than a read-then-write sequence in application code. The design is anchored by the stock invariants Available + Reserved + Confirmed = Total and Reserved + Confirmed ≤ Total, together with an idempotency token that prevents duplicate orders when a buyer retries after a dropped connection.

The resulting artefacts show a coherent path from requirements to behaviour: the use case defines the main and exception flows; the sequence and activity models show duplicate detection, availability checks and atomic reservation; the state model carries the order through Placed → Confirmed → Assigned → In Transit → Delivered with explicit Declined, Cancelled and Reservation Expired outcomes; and the domain model assigns stock, lifecycle, history and reporting responsibilities to stable concepts. Acceptance criteria AC-01 to AC-05 provide directly testable evidence for the core order-placement slice. The supplied Lab 5 package does not contain implementation source code, so implementation and measured test-result claims are deliberately limited to what the supplied artefacts substantiate.

Overall, the Phase 1 design is internally consistent and ready to guide the Phase 2 vertical slice, subject to the documented open issues: finalising the configurable reservation-expiry duration, confirming reporting treatment for exception states, and retaining editable modelling sources alongside exported evidence.

2. Problem evidence, stakeholders and scope

2.1 Approved problem and evidence

The approved project baseline identifies a market-access and coordination problem in Botswana's horticulture value chain. Farmers need reliable buyers before produce deteriorates and greater visibility of demand; smaller buyers need transparent access to local supply, location and price; and transport providers need sufficiently specified delivery jobs. The project responds by coordinating produce listing, ordering and delivery between verified parties.

Evidence basis for this report: the supplied Phase 1 draft states the project goal and approved scope, while the Requirements.md artefact explicitly traces the requirements to the approved project proposal dated 7 August 2026. The Phase 1 draft also records the final project repository and the integration issues addressed during Lab 5.

2.1.1 Evidence and approval trace

The available Phase 1 evidence is primarily project-controlled evidence. The supplied Lab 5 package does not contain an external empirical dataset, field-survey results or production statistics; consequently, this report does not present such material as completed evidence. Instead, the approved problem, requirements, use case, quality scenarios and UML artefacts are used as the evidence base for the design baseline.

Evidence source	What it establishes	Use in this report
Approved project baseline (7 Aug. 2026)	Approved problem, goal and scope	C1 baseline and objectives
Requirements.md	FR-01 to FR-13 and requirement traceability	Functional baseline and links

UC.md / acceptance-criteria.md	Detailed UC-01 behaviour and AC-01 to AC-05	Core use case and acceptance oracles
quality-scenarios.md	QS-01 to QS-10 measurable quality scenarios	Quality obligations and test design
Lab 5 UML artefacts	Use-case, domain, sequence, activity and state evidence	Cross-artefact consistency

2.2 Goal and objectives

Goal: improve market access and reduce avoidable post-harvest losses in Botswana's horticulture value chain by digitally coordinating produce listing, ordering and delivery between verified farmers, buyers and transport providers.

1. By Phase 2, enable a verified farmer to publish a produce listing and a buyer to place an order that the farmer confirms and a transporter delivers end-to-end using synthetic data and automated success-path tests.
2. Guarantee stock integrity so confirmed order quantities never exceed a listing's available quantity, including under concurrent or duplicate order attempts.
3. Enable a buyer to locate available produce by crop and district and complete order placement in five or fewer screens, while allowing an administrator to generate an order-status report from live system data.

2.3 Stakeholders and concerns

Stakeholder	Need / goal	Key concern
Smallholder horticulture farmer	Reliable buyers before produce deteriorates; visibility of demand and fair pricing	Digital literacy, data costs and trust
Buyer - restaurant, small retailer or street trader	Consistent local supply with transparent availability, location and price	Produce quality/grading and fulfilment reliability
Transport provider	Well-specified pickup/delivery jobs	Cancellations after commitment and spoilage responsibility
Platform administrator	Verify users, monitor activity and produce reports	Fraudulent/duplicate listings and dispute workload
Ministry of Agriculture / Botswana Horticulture Council	Sector-level visibility and evidence of local market access	Data accuracy and protection of farmer information

2.4 System boundary and scope

In scope:

- Farmer and buyer registration with administrator verification using simulated documents.
- Produce listings containing crop, grade, unit price, available quantity, district and availability window.
- Listing lifecycle: Active, Sold Out and Expired.
- Order placement and lifecycle: Placed -> Confirmed -> Assigned -> In Transit -> Delivered.
- Atomic stock reservation with explicit Declined, Cancelled and Reservation Expired release rules.
- Transport request and assignment for confirmed orders, plus basic administrative order-status reporting.

Out of scope:

- Digital payments, escrow and financial settlement.

- Warehouse booking and storage management.
- AI recommendations, weather/statistics integrations, predictive analytics and crop-disease reporting.
- Livestock trading, financing and extension-officer services.
- Live third-party integrations; notifications and maps remain stubbed.

2.5 Assumptions and constraints

- Development and demonstration use synthetic names, IDs, contacts and coordinates.
- Buyer and farmer accounts are administrator-verified before they can order or list produce.
- Connectivity may be intermittent, especially in rural contexts; therefore retries must be safe.
- The specific reservation-expiry duration remains an open configuration decision.
- External routing and SMS capabilities are represented by stubs rather than live integrations in the approved scope.

3. Glossary

Term	Definition
Farmer	Registered, administrator-verified smallholder horticulture producer who lists produce for sale.
Buyer	Registered, administrator-verified restaurant, small retailer or street trader who orders produce.
Transporter	Registered delivery provider who accepts and fulfils transport jobs for confirmed orders.
Administrator	Platform operator who verifies registrations, monitors activity and produces reports.
Listing	A farmer-published offer specifying crop, grade, unit price, available quantity, district and availability window.
Listing state	Active (orderable), Sold Out (remaining quantity is zero) or Expired (availability window has passed).
Order	A buyer request for a specific quantity against a specific listing, together with its lifecycle record.
Order state	Placed, Confirmed, Assigned, In Transit or Delivered; Declined, Cancelled and Reservation Expired represent exception paths.
Stock reservation	The atomic decrement of a listing's remaining available quantity performed at order placement.
Verified	Registration status granted by an administrator after checking submitted identification/verification information.
District	Botswana administrative district used to locate listings and filter buyer searches.

Term	Definition
Availability window	Date range during which the listing is available for order.
Transport job	Delivery task created for a confirmed order and accepted by a transporter.
Idempotency token	Client-generated identifier used to recognise a retried order request as the same logical submission.
Order-status report	Administrator-facing summary of order counts by state over a specified date range, derived from live order data.

4. Functional requirements and use cases

4.1 Actors

Actor	Type	Goal
Farmer	Primary human	List produce, confirm orders and get produce delivered.
Buyer	Primary human	Find and order available local produce reliably.
Transporter	Primary human	Receive and fulfil well-specified delivery jobs.
Administrator	Primary human	Verify accounts, monitor activity and produce reports.
OpenRouteService	Secondary external (stubbed)	Supply route information when transport routing is needed.
SMS Gateway	Secondary external (stubbed)	Deliver status notifications.

4.2 System use-case model

The system-level use-case model complements the functional-requirement baseline by making the approved system boundary, primary actors and external service stubs explicit. It covers registration and verification, produce listing and search, order placement and atomic stock reservation, order confirmation/decline, status history, transport coordination, delivery updates, receipt confirmation and administrative reporting. The <<include>> relationship from Place Order to Reserve Stock (atomic) makes the stock-integrity mechanism an essential part of successful order placement; notification and routing remain isolated behind stubbed external systems in Phase 1.

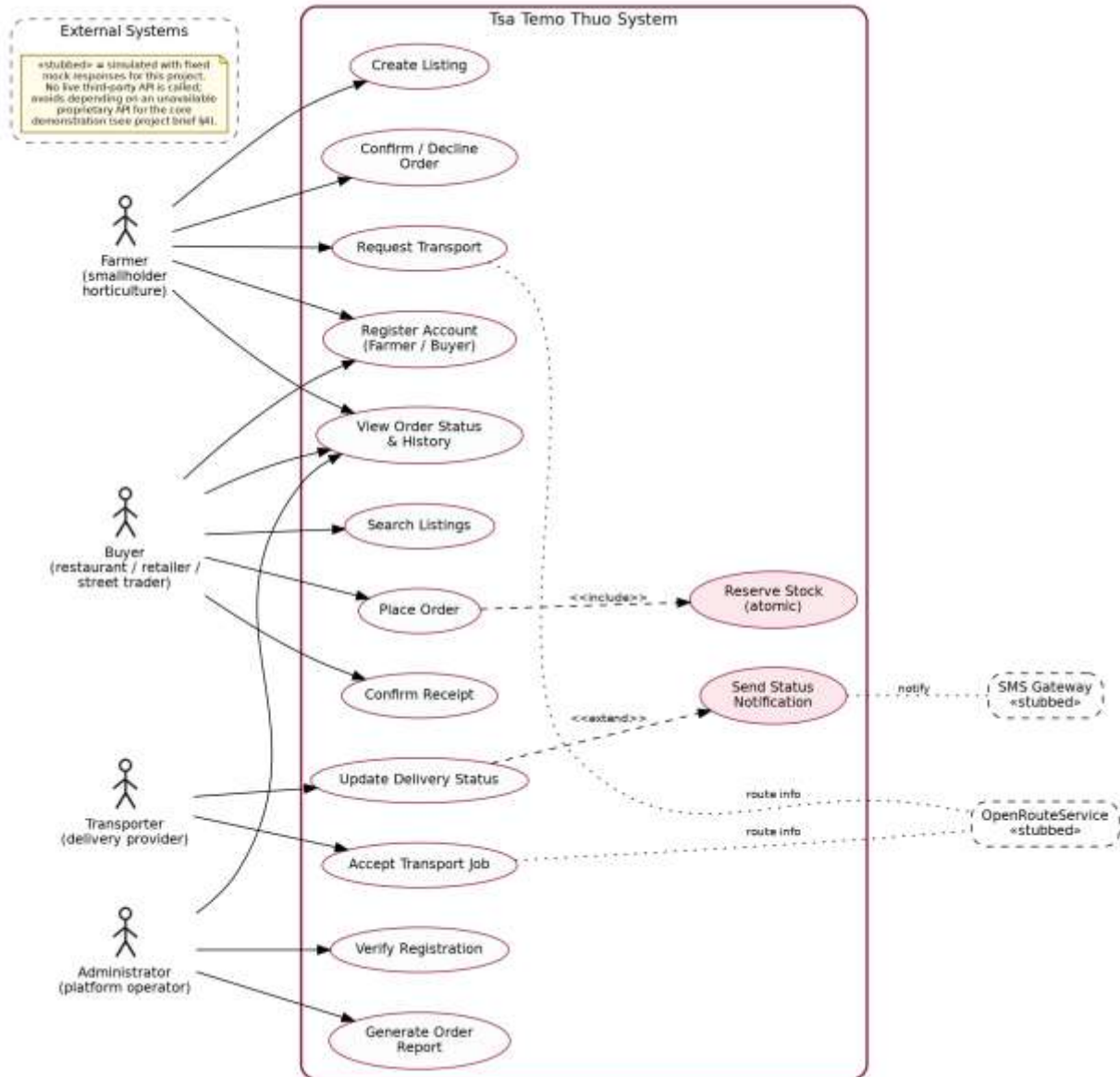


Figure 1. Approved system use-case diagram for Tsa Temo Thuo.

The diagram is consistent with FR-01 to FR-13: registration and verification cover FR-01/FR-02; listing creation and search cover FR-03/FR-04; Place Order and Reserve Stock cover FR-05 to FR-07; Confirm/Decline Order covers FR-08; listing lifecycle and transport interactions cover FR-09 to FR-11; and status/history plus reporting cover FR-12/FR-13. The model therefore provides the visual bridge between actor goals, UC-01 behaviour and the later interaction/lifecycle artefacts.

4.3 Functional requirements baseline

ID	Requirement	Primary actor(s)
FR-01	Allow prospective farmer or buyer to submit registration with identifying details and supporting verification information.	Farmer, Buyer
FR-02	Allow administrator to review pending registration and record Approve or Reject, including a reason when rejected.	Administrator



ID	Requirement	Primary actor(s)
FR-03	Allow verified farmer to create a listing specifying crop, grade, unit price, quantity, district and availability window; state becomes Active.	Farmer
FR-04	Allow verified buyer to search Active listings filtered at minimum by crop type and district.	Buyer
FR-05	Allow verified buyer to place an order for a quantity not exceeding remaining stock on an Active listing.	Buyer
FR-06	Atomically reserve ordered quantity so Placed + Confirmed quantities never exceed original available quantity, including concurrency and duplicate attempts.	System
FR-07	Reject quantity-over-availability and Sold Out/Expired orders without changing reserved stock.	System
FR-08	Allow listing owner to Confirm or Decline a Placed order; Decline releases reservation.	Farmer
FR-09	Automatically transition listing to Sold Out at zero remaining quantity and Expired after the availability window; reject orders against either state.	System
FR-10	Allow farmer to request transport for a Confirmed order and available transporter to accept it, producing Assigned state.	Farmer, Transporter
FR-11	Allow assigned transporter to update delivery through In Transit and Delivered; buyer can confirm receipt.	Transporter, Buyer
FR-12	Allow authorised farmer, buyer and administrator to view current order state and full status history.	Farmer, Buyer, Administrator

ID	Requirement	Primary actor(s)
FR-13	Allow administrator to generate an order-activity report by state over a date range from live data.	Administrator

4.4 Core use case - UC-01 Place Order

Primary actor: Buyer (restaurant, small retailer or street trader). Secondary actors: Farmer, who owns the listing and is notified; and the System, which validates the order and atomically manages stock.

Preconditions: the Buyer is registered and Verified; an Active listing exists; the listing has remaining quantity greater than zero; the availability window has not passed; and the buyer submits a positive quantity against a listing containing the required crop, grade, price, quantity, district and availability information.

Success guarantee: an Order exists in Placed state, linked to the buyer and listing; ordered quantity has moved from Available to Reserved exactly once; the total quantity is unchanged; the operation is atomic; order timestamp and relationships are persisted; and the farmer can see the resulting order.

4.5 Main success scenario

1. Buyer searches Active listings and selects a suitable listing.
2. Buyer reviews crop, grade, unit price, district, availability window and remaining quantity.
3. Buyer enters desired quantity and submits with a client-generated idempotency token.
4. System checks whether the token already exists; if it does, the existing order state is returned.
5. System verifies the listing remains Active and validates the requested quantity against current remaining stock.
6. System performs the atomic stock reservation.
7. System creates the Order in Placed state and persists the buyer/listing relationship and timestamp.
8. System appends a Placed OrderStatusEvent for history and auditability.
9. If remaining quantity reaches zero, the Listing becomes Sold Out.
10. System notifies the Farmer and returns a Placed confirmation to the Buyer.

4.6 Exception and alternative flows

Condition	System response	Invariant / result
Requested quantity exceeds availability	Reject; create no valid reservation; return actual maximum available quantity.	Available stock unchanged.
Listing is Sold Out or Expired	Reject order and return to Search Listings.	No reservation created.
Concurrent orders compete for last stock	Atomic reservation determines the winner; losing requests are rejected through the same validation path.	No oversell; no negative stock.
Retry after dropped connection	Same idempotency token is recognised; original Order is returned.	No second Order; no second reservation.
Farmer declines	Order becomes Declined and reservation is released once.	Released quantity returns to Available.

Condition	System response	Invariant / result
Buyer cancels a Placed order where authorised	Order becomes Cancelled and reservation is released.	Cancelled order no longer holds stock.
Reservation expires	Order becomes Reservation Expired and reserved stock is released, subject to listing availability window.	Released stock becomes reusable only while listing can still be ordered.
Farmer confirms	Order becomes Confirmed; Reserved transfers to Confirmed.	Available is not deducted a second time.

4.7 Acceptance criteria

ID	Scenario	Expected result
AC-01	50 kg remaining; buyer orders 20 kg.	Order Placed; remaining quantity 30 kg.
AC-02	10 kg remaining; buyer requests 15 kg.	Request rejected; no Order; stock remains 10 kg; maximum available shown as 10 kg.
AC-03	Listing Sold Out or Expired.	Rejected; no reservation made.
AC-04	Two simultaneous 5 kg requests against 5 kg remaining.	Exactly one request succeeds; listing becomes Sold Out; other request rejected; no oversell state.
AC-05	Same order request retried using the same idempotency token.	Original Order returned; no duplicate Order or reservation.

5. Quality requirements

The project quality requirements are expressed as measurable stimulus-context-response-measure scenarios. These are not generic non-functional statements; each one is tied to a project risk, requirement or explicit objective.

ID	Quality concern	Measure
QS-01	Data integrity under concurrent orders	0 oversell states across 100 simulated concurrent-order test runs.
QS-02	Listing search performance	95th-percentile server-side response time under 2 seconds across 50 repeated searches on 500 synthetic Active listings.
QS-03	Dropped connectivity / duplicate prevention	0 duplicate orders and 0 duplicate reservations across 20 simulated

ID	Quality concern	Measure
		drop-and-retry runs using the same idempotency token.
QS-04	Usability of order placement	At least 4 of 5 synthetic first-time users complete the find-and-order task in 5 or fewer screens.
QS-05	Authorisation of farmer confirmation/rejection	100% rejection of unauthorised confirmation/rejection attempts, with 0 successful unauthorised state changes.
QS-06	Stock restoration after rejected/cancelled/expired orders	100% of tested release paths restore reserved quantity correctly; 0 permanently locked stock cases.
QS-07	No double deduction on confirmation	100% of confirmation tests preserve Available + Reserved + Confirmed = Total with 0 double deductions.
QS-08	Produce-quality information responsibility	100% of tested listings show farmer-declared grade before ordering; every quality complaint is traceable to farmer, listing and order.
QS-09	Recoverability of order history after failure	100% of order records preserve pre-failure state and history across 10 simulated crash/restart runs.
QS-10	Stock consistency across full lifecycle	100% of lifecycle tests preserve both stock equations with 0 negative-stock, oversell or double-deduction states.

5.1 Resulting design obligations

- Implement stock reservation as an atomic data-layer operation rather than a read-then-write sequence.
- Use a unique idempotency-token boundary for order submission.
- Persist order state changes as status events so full history survives normal operations and restart recovery.
- Enforce owner-based authorisation for confirmation and rejection.
- Keep the buyer flow to five or fewer screens and expose only the information needed for a confident order.
- Use synthetic data for development and demonstration and keep third-party notifications/routing stubbed in the approved scope.

6. Analysis/domain model and responsibilities

The supplied domain model expresses a coherent set of responsibilities around Account, Farmer, Buyer, Administrator, Listing, Order, TransportJob, OrderActivityReport and OrderStatusEvent-related history. The key analytical decision is to keep the stock lifecycle on Listing and the order lifecycle on Order, while treating status history as persistent evidence rather than deriving history from current state alone.

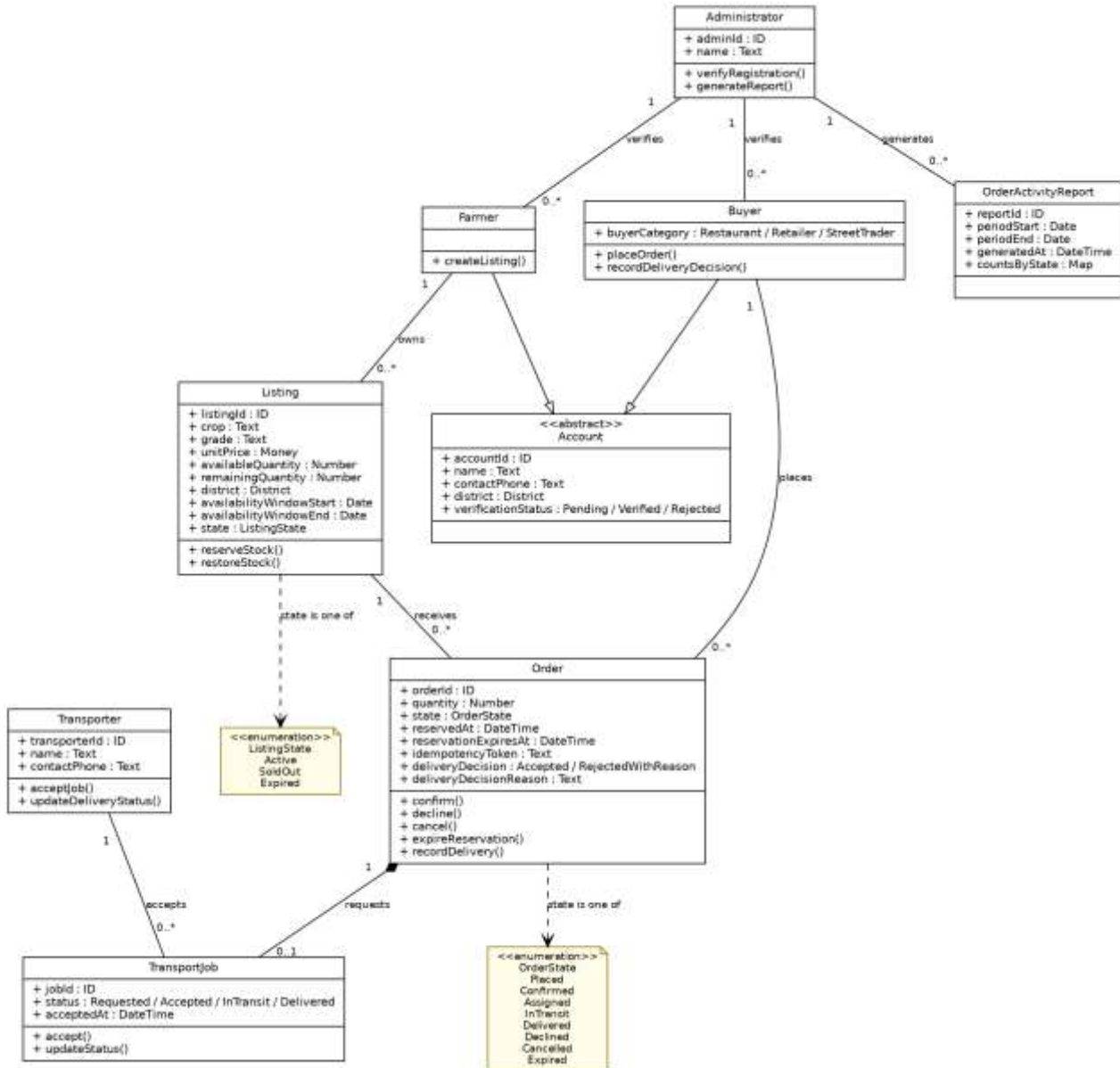


Figure 2. Phase 1 domain model

6.1 Business rules

- Available + Reserved + Confirmed = Total.
- Reserved + Confirmed ≤ Total.
- A successful order moves quantity from Available to Reserved exactly once.
- Farmer confirmation transfers Reserved to Confirmed; it does not deduct Available again.
- Decline, cancellation and reservation expiry return reserved quantity to Available exactly once.
- If a decline restores quantity before the availability window ends, a previously Sold Out listing may return to Active.
- Only the owner of a listing may confirm or decline its orders.
- Expired reservations cannot be reused after the listing availability window has ended.



- The system records farmer-declared produce grade/quality; it does not independently certify physical quality.

6.2 Responsibility allocation

Concept	Primary responsibility	Why this allocation is appropriate
Listing	Own quantity state; reserveStock(); restoreStock(); lifecycle state.	The stock being protected belongs to the listing, so the consistency boundary is located with the stock-bearing entity.
Order	Own lifecycle state; link buyer/listing; confirm(), decline(), cancel(), expireReservation(), recordDelivery().	Order is the state-sensitive transaction whose history must remain auditable.
OrderStatusEvent	Persist state transitions with timestamps.	Makes history explicit and supports FR-12 and reporting/audit requirements.
Tsa Temo Thuo System / application orchestration	Validate preconditions, coordinate atomic reservation, create order, append event and notify farmer.	Coordinates the use case without moving stock rules into a procedural read-then-write workflow.
Administrator	verifyRegistration(); generateReport().	Matches approved verification and reporting responsibilities.
TransportJob	Create/request transport and manage acceptance/status.	Keeps delivery coordination separate from core order quantity rules.

7. Interaction and lifecycle

7.1 Sequence model

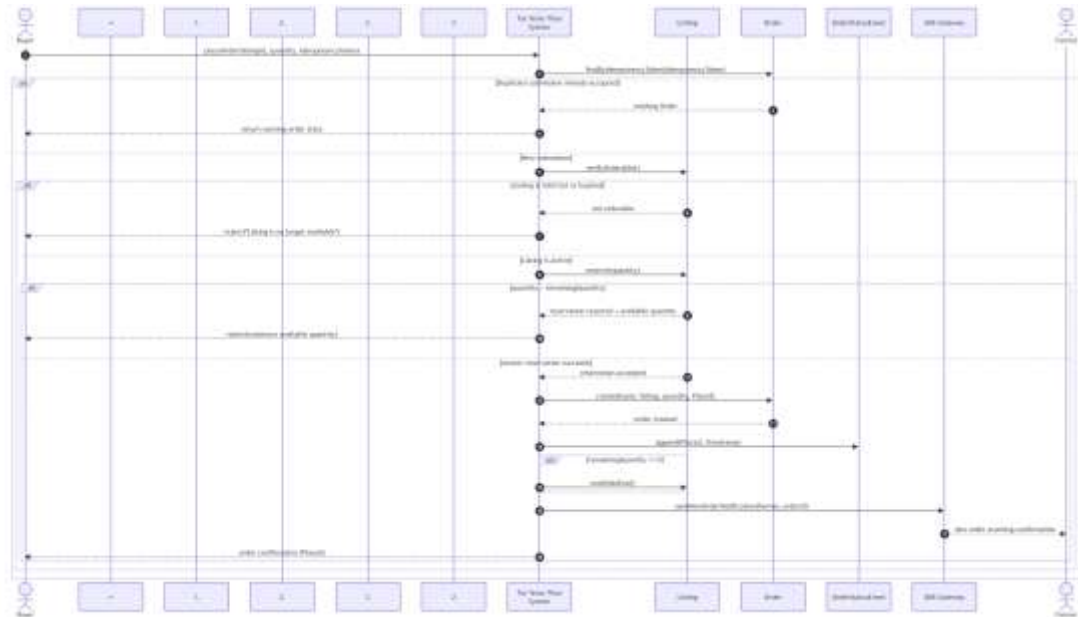


Figure 3. UC-01 sequence model showing duplicate detection, atomic reservation, order creation, history and notification.

The sequence model makes the consistency boundary explicit. The Buyer sends one placeOrder request with an idempotency token. The system first checks for an existing token; for a new request it verifies that the Listing is still orderable, asks the Listing to perform the reservation atomically, creates the Order only after reservation succeeds, persists the status event and optionally marks the listing Sold Out when no quantity remains. Notification is sent through the stubbed SMS Gateway.

7.2 Activity model

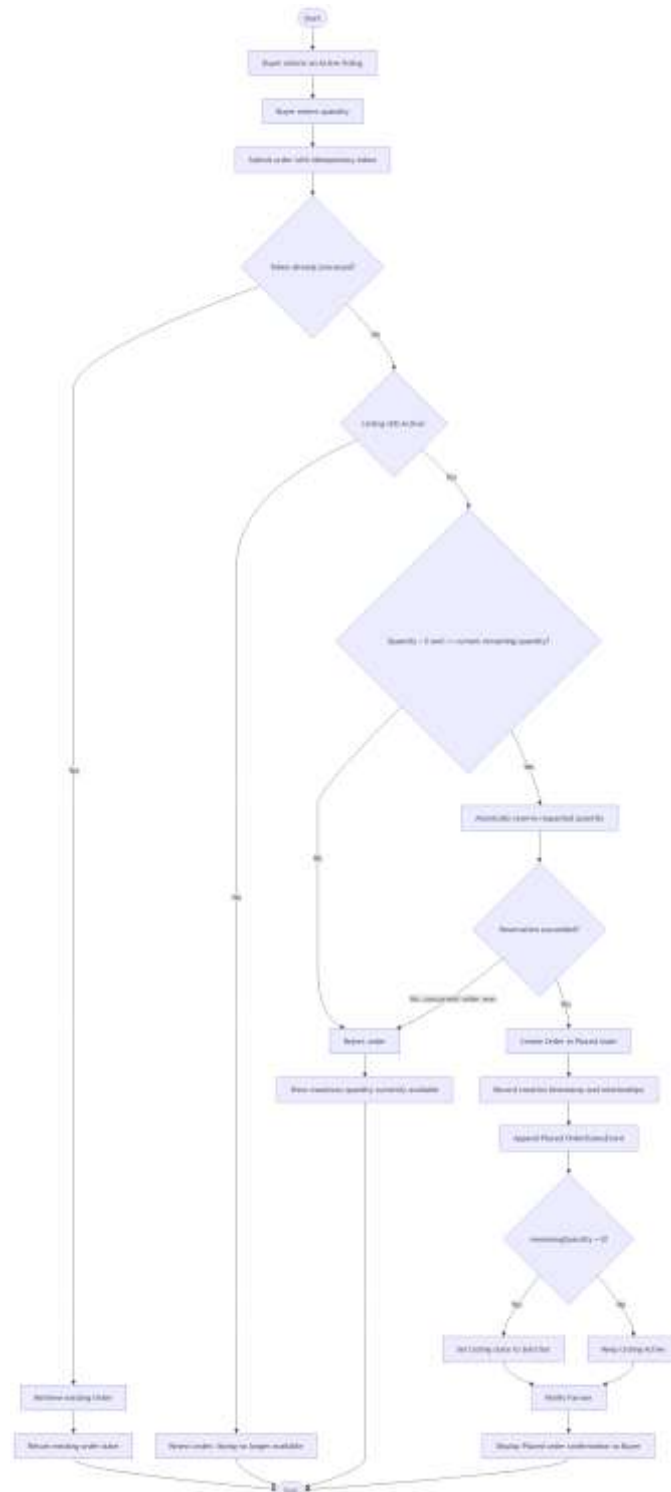


Figure 4. UC-01 activity model showing duplicate, availability, quantity and atomic-reservation decisions.

The activity model is consistent with the sequence model because it represents the same control decisions at workflow level: duplicate token detection, Active-listing validation, positive-and-within-stock quantity validation, atomic reservation success/failure, Sold Out transition, farmer notification and buyer confirmation.

7.3 State/lifecycle model

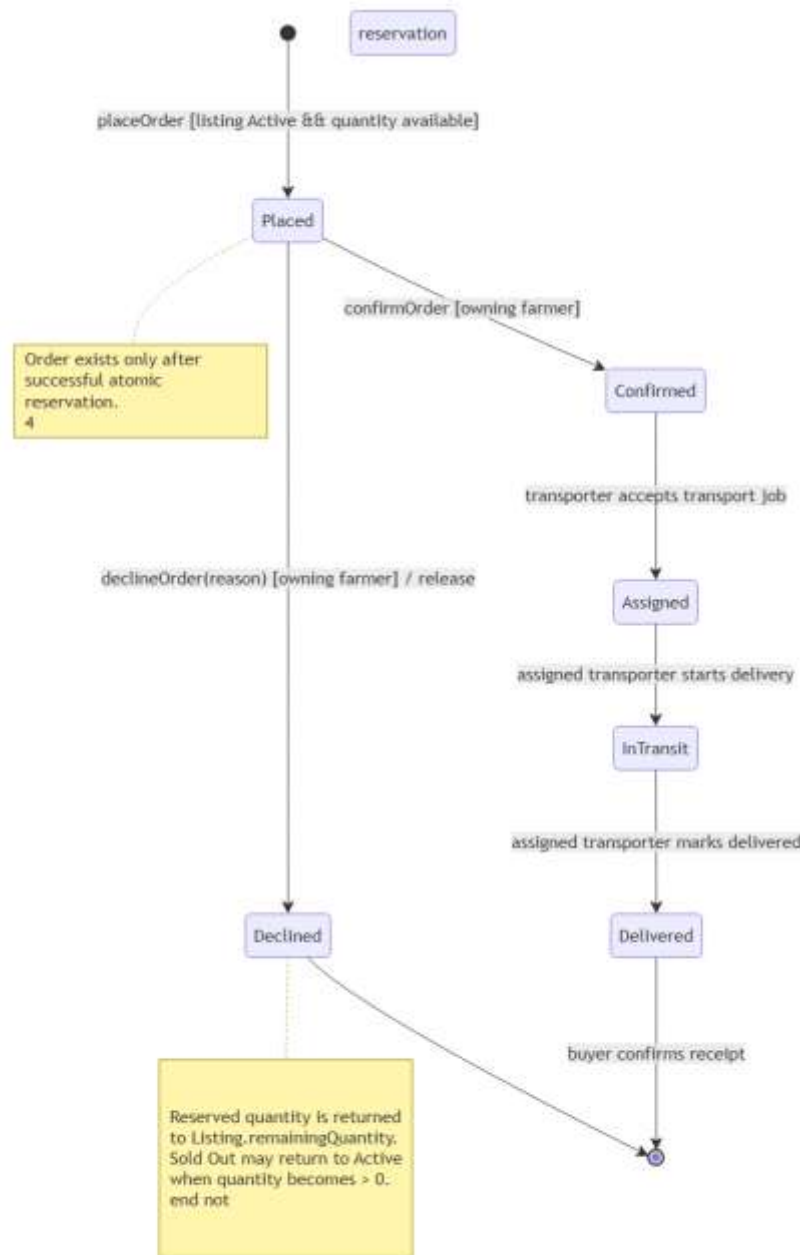


Figure 5. Order lifecycle state machine

The approved lifecycle specification is Placed -> Confirmed -> Assigned -> In Transit -> Delivered, with Declined, Cancelled and Reservation Expired specified as exception outcomes. The supplied state-machine export explicitly visualises Declined but does not currently show separate Cancelled and Reservation Expired states; those two outcomes are therefore retained as controlled model-update items for Phase 2. The key lifecycle invariant is that an order exists in Placed only after successful reservation; confirming transfers reserved quantity to confirmed quantity; declining releases the reservation; and delivery progression changes order state without reusing the stock reservation rule.

7.4 Consistency across models

Concern	Use case	Sequence/activity	State/domain result
Duplicate request	Retry returns existing Order.	findByIdempotencyToken() precedes new processing.	No duplicate Order; no duplicate reservation.

Concern	Use case	Sequence/activity	State/domain result
Unavailable listing	Sold Out/Expired rejected.	verifyOrderable() blocks reservation.	Listing unchanged; order not created.
Concurrent last-stock requests	One controlled success, remaining requests rejected.	Atomic reserve determines winner.	Stock invariants maintained.
Successful order	Reservation then Placed order.	reserve -> create -> append event.	Placed; Reserved increases by exactly ordered quantity.
Confirmation	Reserved -> Confirmed.	Lifecycle continuation beyond core slice.	No second Available deduction.
Decline/cancel/expiry	Release reserved quantity once.	Lifecycle continuation.	Available restored subject to availability window.

8. Architecture and decisions

Because the supplied Phase 1 package does not contain application source code, this section documents the approved conceptual architecture implied by the domain, interaction and quality artefacts rather than claiming an implemented technology stack.

8.1 Architecture drivers

- Strong stock integrity under concurrent requests.
- Safe retry behaviour under intermittent connectivity.
- Clear ownership of lifecycle transitions and authorisation.
- Persistent, queryable status history for operational monitoring and reporting.
- A simple buyer workflow suitable for first-time users and rural connectivity constraints.
- Stubbed external dependencies so the core vertical slice can be tested deterministically.

8.2.1 Architecture alternatives considered

Alternative	Strength	Main weakness against drivers	Decision
Atomic reservation at persistence boundary	Strong concurrency correctness and clear invariant boundary.	Requires transactional coordination.	Selected
Read-then-write in application service	Simple and familiar implementation.	Race condition can oversell the last stock.	Rejected
Create Order first, reserve later	Separates order creation from inventory logic.	Failure between operations can create an unbacked order.	Rejected
UI-only duplicate prevention	Cheap for normal repeated clicks.	Does not protect against retries or multiple clients.	Rejected

8.2 Conceptual component model

Component	Responsibility	Key dependency
Buyer-facing UI	Search listings; show details; accept quantity; submit idempotent order request; show result.	Application/API layer



Component	Responsibility	Key dependency
Application / use-case service	Validate request, check idempotency, coordinate reservation and order creation, emit status event and notification.	Listing, Order, persistence, notification stub
Listing domain component	Enforce Active/Sold Out/Expired rules and atomic quantity reservation/release.	Persistent data store / transaction boundary
Order domain component	Enforce Placed/Confirmed/... lifecycle and ownership rules.	Status history and transport coordination
Persistence layer	Store listings, orders, status events, registrations and report source data transactionally.	Database
Notification adapter (stub)	Send order notification without coupling core workflow to a live SMS provider.	SMS Gateway stub
Routing adapter (stub)	Provide route information when delivery coordination reaches routing concern.	OpenRouteService stub

8.3 Architecture decision records

ADR	Decision	Rejected alternative	Consequence
ADR-01	Reserve stock atomically at the data layer.	Read available quantity, then write the new quantity in separate steps.	Adds transactional/concurrency complexity but establishes a reliable zero-oversell boundary.
ADR-02	Use idempotency tokens for order submission.	Rely on the UI to prevent repeat clicks or simply create another order on retry.	Requires token storage/uniqueness but makes network retries safe.
ADR-03	Persist status events.	Infer history only from the current Order state.	Adds persistence and query cost but supports FR-12, auditability and reporting.
ADR-04	Keep SMS and routing integrations stubbed in Phase 1.	Couple the vertical slice to live external services.	Improves deterministic testing and keeps approved scope controlled.

9. Data, API/service and deployment design

9.1 Logical data model

The supplied domain model supports the following logical records: Account (with specialised Farmer/Buyer roles), Listing, Order, TransportJob, OrderStatusEvent/history and OrderActivityReport. Listing carries crop, grade, price, quantity, district, availability dates and lifecycle state. Order carries quantity, state, reservation timestamps, idempotency token and delivery decision information. The design should persist the stock-related quantities and transitions atomically so that lifecycle history can be reconciled to the listing total.

9.2 Service contracts

Service / operation	Input	Success	Failure / invariant
placeOrder	listingId, buyerId, quantity, idempotencyToken	Placed order + reservation + event.	Reject unavailable/invalid/duplicate requests; no duplicate reservation.
findByIdempotencyToken	idempotencyToken	Return existing Order/state.	No new Order created.
reserveStock	listingId, quantity	Atomic Available -> Reserved movement.	No partial reservation; no negative Available.
confirmOrder	orderId, owning farmer	Order Confirmed; Reserved -> Confirmed.	Reject unauthorised actor; no second Available deduction.
declineOrder	orderId, owning farmer, reason	Order Declined; reservation released.	Reject unauthorised actor; release exactly once.
generateOrderActivityReport	date range	Counts by order state from live data.	No stale snapshot semantics claimed.

9.3 Deployment and failure behaviour

The exact deployment topology is not evidenced in the supplied package. At the design level, the core invariant requires a transactional persistent store behind the application service. Failure during order placement must not leave an Order without the stock it represents; therefore reservation and Order creation are treated as one successful transaction boundary. For crash recovery after a Confirmed state, QS-09 requires the persisted Order state and status history to match the pre-failure state exactly.

External notification and routing are intentionally stubbed. This means failure of SMS or route retrieval must not redefine stock integrity or order state; the core transaction remains the authoritative business record.

10. Detailed design, patterns and framework

10.1 Interfaces and responsibilities

Interface / service abstraction	Purpose
IOrderService	Orchestrates UC-01 and subsequent order lifecycle transitions.
IListingService / Listing repository	Searches listings and provides the atomic stock reservation/release boundary.
IOrderRepository	Creates and retrieves Orders and enforces idempotency-token lookup.
IOrderStatusEventRepository	Appends persistent lifecycle events and exposes history.
INotificationGateway	Abstracts SMS notification; Phase 1 implementation is a stub.
IRoutingService	Abstracts route information; Phase 1 implementation is a stub.

10.2 Design patterns

- Service/orchestration layer for coordinating the use case while leaving business invariants with domain responsibilities.
- Transactional unit-of-work boundary for atomic stock reservation plus order creation.
- Repository abstraction around persistent Orders, Listings and status events.
- Adapter pattern for SMS and routing stubs to prevent external providers from leaking into core domain logic.
- Idempotent command handling for retry-safe order submission.

10.3 Rejected alternatives

Alternative	Why rejected
Read-then-write stock update	Cannot safely arbitrate simultaneous buyers observing the same stock.
Create Order first, reserve later	Failure between operations can create an order without stock backing.
UI-only duplicate prevention	Cannot protect against network retries, repeated API submissions or concurrent clients.
History inferred from current state	Loses the evidence needed for a full status history and reliable audit/reporting.

10.4 Framework boundary

No concrete framework or programming-language implementation is claimed in this Phase 1 package. The report therefore specifies behavioural and architectural contracts without attributing them to a particular web framework, ORM or database product.

11. User-interface, accessibility and usability

11.1 Buyer task flow

1. Open available produce search.
2. Filter by crop and district.
3. Select a listing and review crop, grade, price, district, availability window and quantity.
4. Enter a valid quantity and submit.
5. Receive a Placed confirmation or an actionable validation/error message.

11.2 Alternative UI states

State	Required user-facing response
No matching listings	State that no matching Active listing is available; avoid presenting expired/sold-out entries as orderable.
Quantity too high	Show actual available quantity and allow correction without losing the selected listing.
Listing becomes unavailable before submit	Explain that the listing is no longer available and return to the search context.
Concurrent request loses reservation	Explain that requested quantity is no longer available rather than showing a generic failure.
Retry after connection loss	Return the existing order state rather than prompting a new order.

11.3 Accessibility and usability obligations

- Keep the primary ordering task within five screens.
- Use clear field labels for quantity, crop, district and grade; avoid relying on colour alone to communicate state.
- Provide actionable validation messages, including the actual maximum quantity where applicable.
- Keep interaction targets usable on mobile-sized screens and reduce unnecessary navigation for first-time users.
- Present listing information before order submission so the buyer can make an informed choice.

The supplied package contains no completed UI wireframes or usability-test recordings. Accordingly, the report treats the items above as design obligations derived from Objective 3 and QS-04 rather than claiming completed user testing.

12. Implementation vertical slice

Approved core use case: UC-01 Place Order. The intended Phase 2 vertical slice is the end-to-end path from verified buyer selection of an Active listing through validated quantity submission, atomic reservation, Placed order creation, status history recording, notification and buyer confirmation, with AC-01 to AC-05 covering success, validation, lifecycle rule, concurrency and duplicate-retry behaviour.

12.1 Repository evidence

Path	Evidence in supplied package
README.md	Project overview, Team 10 identity, core feature and FR-05/FR-06/FR-07 focus.
docs/Requirements.md	Approved functional requirements and traceability.
docs/acceptance-criteria.md	AC-01 to AC-05 for the vertical slice.
docs/quality-scenarios.md	QS-01 to QS-10 measurable quality scenarios.
docs/Glossary.md	Stable domain terminology.
docs/use cases/UC.md	Detailed UC-01 main and alternative flows.
docs/domain-model.png	Domain analysis model.
models/Sequence diagram.drawio.png	UC-01 interaction model.
models/activity.drawio.png	UC-01 activity model.
models/state machine diagram.png	Order lifecycle model.
submissions/Tsa Temo Thuo Phase1 draft.pdf	Integrated Phase 1 draft and checklist.

12.2 Setup and run status

The README explicitly states that development setup instructions are still to be added. The supplied Lab 5 archive contains documentation and model evidence but no application source tree. Consequently, this report does not fabricate build commands, runtime prerequisites, endpoints or deployment steps. Those details become mandatory release artefacts for the implementation phase.

12.2.1 Reproducibility checklist

Reproducibility item	Phase 1 evidence	Status / next action
Stable project identity	Team 10, repository URL and submission tag recorded in the report.	Complete
Editable report	Submission is maintained as an editable DOCX.	Complete
Editable UML source	Supplied archive contains exported PNG models; editable Draw.io sources are not included.	Pending: retain/source editable model files for Phase 2
Setup/run instructions	README states setup instructions still need to be added.	Pending implementation phase
Build/test evidence	No application source or executed automated test logs are in the supplied Lab 5 package.	Pending implementation phase
Artefact-to-repository consistency	Report records the repository and supplied artefacts used.	Complete for Phase 1 design evidence

13. Testing and results

13.1 Requirement-linked test design

Test ID	Trace	Purpose	Expected evidence
T-AC01	FR-05, AC-01	Normal order placement.	Placed order; stock reduced by exact order quantity.
T-AC02	FR-06/07, AC-02	Quantity exceeds remaining stock.	Rejected; no Order; stock unchanged; actual maximum shown.
T-AC03	FR-07/09, AC-03	Sold Out / Expired listing.	Rejected; no reservation.
T-AC04	FR-06, AC-04, QS-01	Concurrent orders for final stock.	Exactly one accepted; 0 oversell states.
T-AC05	FR-06, AC-05, QS-03	Dropped-connection retry.	Original order returned; no duplicate order/reservation.
T-QS05	FR-08, QS-05	Unauthorised confirmation/rejection.	100% rejection of unauthorised attempts.
T-QS06/07	FR-08/09, QS-06/07/10	Lifecycle stock release and confirmation transfer.	Invariants preserved and no double deduction.
T-QS09	FR-12/13, QS-09	Crash/restart recovery.	Order state/history preserved across restart runs.

13.2 Test oracle - stock invariants

The primary test oracle is not simply the HTTP response; it is the stock state before and after each operation. Every successful or released quantity must reconcile to the equations $\text{Available} + \text{Reserved} + \text{Confirmed} = \text{Total}$ and $\text{Reserved} + \text{Confirmed} \leq \text{Total}$. The acceptance criteria and quality scenarios are therefore sufficient to define objective pass/fail rules even before a concrete implementation is attached.

13.3 Results status

The supplied package defines the tests and measures but does not include executed test logs, automated test source or measured runtime output. Therefore the appropriate status for this Phase 1 report is "test design specified; execution evidence pending implementation" rather than claiming passes that are not evidenced in the submission package.

14. Reconciliation and change history

The traceability chain is intentionally bidirectional: requirements are realised in the use case and models, while acceptance and quality evidence act as objective test oracles for those same design decisions. This reduces the risk that one diagram evolves independently of the requirement baseline.

14.1 Model-to-requirement-to-test mapping

Requirement / rule	Use case evidence	Model evidence	Acceptance / quality evidence
FR-05 Order placement	UC-01 main flow	Sequence + activity	AC-01
FR-06 Atomic reservation	UC-01 special requirements	Sequence + domain model	AC-04, AC-05, QS-01, QS-03
FR-07 Invalid/unavailable order	UC-01 extensions	Activity decisions	AC-02, AC-03
FR-08 Farmer decision	UC-01 extensions	State model	QS-05, QS-06
FR-09 Listing lifecycle	Preconditions + alternative flows	State/domain lifecycle	AC-03, QS-06, QS-10
FR-10/FR-11 Transport/delivery	Lifecycle continuation	State model + TransportJob domain concept	Phase 2 end-to-end objective
FR-12 Status history	Persist Placed event + lifecycle	OrderStatusEvent concept	QS-09
FR-13 Reporting	Domain report concept	Administrator -> OrderActivityReport	Reporting scope; execution pending implementation

14.2 Focused review, revision and issue closure

The integrated review changes recorded below are the concrete revisions resulting from cross-artefact checking:

- The lifecycle specification was refined so Declined, Cancelled and Reservation Expired release paths are explicit in the requirements and narrative; the supplied state-machine export still needs separate visual states for Cancelled and Reservation Expired.
- The sequence model was refined to distinguish the System, Listing, Order, OrderStatusEvent, SMS Gateway and human actors.
- The state model explicitly records that an Order exists only after successful atomic reservation, preventing an unbacked Placed order.
- The final Phase 1 checklist identifies remaining decisions rather than silently assuming a reservation-expiry duration or reporting convention.

The Phase 1 review concentrated on the highest-risk correctness boundaries: overselling under concurrency, duplicate submission after connectivity loss, lifecycle release paths, owner authorisation, and consistency between requirements and models. The issue register below distinguishes items resolved through the integrated design from decisions deliberately left as controlled Phase 2 actions.

Review issue	Revision / decision	Closure evidence	Status
Overselling risk was underspecified	FR-06 and the stock equations were tightened around atomic reservation.	UC-01, sequence/activity models, ADR-01, AC-04, QS-01.	Closed

Duplicate retry could create a second order	Idempotency token made part of order submission and lookup.	UC-01 step 4, ADR-02, AC-05, QS-03.	Closed
Release paths could disappear from lifecycle views	Declined, Cancelled and Reservation Expired were made explicit in the requirements/narrative; the exported state model must still be updated to visualise Cancelled and Reservation Expired as separate states.	Requirements/UC-01, business rules, state narrative, QS-06/QS-10; state diagram export currently shows Declined explicitly.	Partially closed / open
Confirmation could double-deduct stock	Confirmation explicitly transfers Reserved -> Confirmed.	Business rules, consistency table, QS-07.	Closed
Reservation expiry policy not final	Duration remains configurable and is a sign-off decision.	Assumptions, risks, release recommendation.	Open / controlled
Exception-state reporting convention not final	Treatment of exception states flagged for Phase 2 clarification.	Risk register and reporting scope.	Open / controlled

15. Risks, technical debt and release recommendation

Risk / debt	Impact	Priority	Mitigation / release condition
Atomic reservation not implemented at database/transaction boundary	Overselling and corrupted stock state	Critical	Do not release the vertical slice until AC-04 and QS-01 pass with concurrency evidence.
Idempotency token not persisted/uniquely enforced	Duplicate orders and duplicate stock reservations after retries	Critical	Do not release until AC-05 and QS-03 pass.
Reservation-expiry duration unresolved	Inconsistent operational behaviour	High	Make duration configurable and record final policy before implementation sign-off.
Exception-state reporting not fully specified	Ambiguous administrator reports	Medium	Confirm treatment of Declined, Cancelled and Reservation Expired orders.
No implementation evidence in Lab 5 package	Cannot yet establish runtime or deployment readiness	High	Attach source, build/run instructions and automated test logs in Phase 2.

Risk / debt	Impact	Priority	Mitigation / release condition
Third-party integrations remain stubbed	Operational notifications/routing not production-ready	Medium	Keep adapters isolated; integrate only after core transaction behaviour is stable.
Produce quality is farmer-declared	Buyer disputes may arise	Medium	Display grade clearly, preserve farmer/listing/order traceability, and avoid implying independent certification.
Digital literacy/connectivity constraints	Usability failure for intended users	Medium	Keep five-screen target, concise validation and retry-safe submission.

15.1 Release recommendation

Phase 1 is suitable as a design baseline, not as a production release. The artefacts form a coherent specification for the vertical slice, and the most important correctness properties are measurable. Release into implementation should proceed only after the critical stock-integrity and duplicate-prevention tests are automated and passed, followed by evidence for authorisation, lifecycle restoration, recovery and the five-screen usability objective.

16. Academic integrity and AI-use record

AI was used as a support tool for brainstorming project and design ideas and for checking the completed report against the requirements provided for CSI473. We supplied the project facts, requirements, models and decisions and the AI suggestions were reviewed by the team and were not treated as evidence of implementation, testing, approval or project ownership. The final report content and technical decisions remained our responsibility as a team.

Purpose	Prompt	How the output was used
Brainstorming	“Help us brainstorm practical features, stakeholders, risks and design considerations for our project based on the requirements we have provided.”	Used to generate and compare ideas; the team selected only ideas consistent with the approved project scope.
Requirement check	“Check this project report against the CSI473 report requirements and identify sections that are missing, weak, inconsistent or unsupported by the supplied project artefacts.”	Used as a checklist to improve completeness and consistency before submission.
Design review	“Review our use cases, quality requirements and UML models for internal consistency. Point out	Used to identify issues for team review and correction in the project artefacts.



	contradictions or missing links without inventing project facts.”	
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16.1 Communication and evidence conventions

The report distinguishes approved facts, design decisions, measurable requirements and implementation status. Runtime claims are not presented as completed evidence where the supplied package contains no execution record. Figures are captioned and referenced in context, and stable project terminology follows the supplied glossary.

References

- [1] Tsa Temo Thuo Team 10, "Project Requirements - Tsa Temo Thuo," docs/Requirements.md, CSI473, 14 Aug. 2026.
- [2] Tsa Temo Thuo Team 10, "Acceptance Criteria - UC-01: Place Order," docs/acceptance-criteria.md, CSI473, 14 Aug. 2026.
- [3] Tsa Temo Thuo Team 10, "Quality Scenarios - Tsa Temo Thuo," docs/quality-scenarios.md, CSI473, 14 Aug. 2026.
- [4] Tsa Temo Thuo Team 10, "UC-01: Place Order," docs/use cases/UC.md, CSI473, 14 Aug. 2026.
- [5] Tsa Temo Thuo Team 10, "Glossary - Tsa Temo Thuo," docs/Glossary.md, CSI473.
- [6] Tsa Temo Thuo Team 10, "CSI473 Phase 1 Draft - Tsa Temo Thuo," submitted 4 Sep. 2026.
- [7] Tsa Temo Thuo Team 10, "Tsa Temo Thuo" README, repository: github.com/Shawnjembu/Tsa-Temo-Thuo.

Appendices

Appendix A - Marking-criteria coverage

This table is a navigation aid showing where each criterion is evidenced in the report; it does not replace the underlying artefacts.

Criterion	Primary evidence in report	Coverage status
C1 Approved problem, evidence, stakeholders and scope	Sections 2.1-2.5; evidence/approval trace; stakeholder table; scope and constraints.	Substantive
C2 Functional requirements, actors, use cases and acceptance	Section 4; actor table; approved use-case diagram; UC-01; AC-01 to AC-05.	Substantive
C3 Measurable quality requirements	Section 5; QS-01 to QS-10; resulting design obligations.	Substantive
C4 Domain model, business rules and responsibilities	Section 6; domain model; invariants; responsibility allocation.	Substantive
C5 Interaction and lifecycle/workflow modelling	Section 7; sequence, activity and state models; consistency table.	Substantive
C6 Traceability, consistency and rationale	Sections 8-10 and 14; requirement/model/test mapping; ADRs.	Substantive



C7 Focused review, revision and issue closure	Section 14.2; issue-closure register; revision record.	Substantive with controlled open items
C8 Reproducible submission and editable evidence	Section 12.2.1; repository evidence; explicit pending source/setup items.	Design-level complete; executable evidence pending
C9 Architecture drivers and realistic alternatives	Sections 8.1-8.4; alternatives and ADRs.	Substantive
C10 Report communication, references and integrity	Section 16.1; Section 16; References; evidence register.	Substantive

Appendix B - Traceability matrix

Objective	Requirements	Acceptance / quality	Primary artefacts
Obj. 1 - end-to-end workflow	FR-01/02/03/05/08/10/11/12	AC-01; QS-06/07/09	Requirements; UC-01; lifecycle/state model; domain model
Obj. 2 - zero oversell / stock integrity	FR-05/06/07/09	AC-02/03/04/05; QS-01/03/06/07/10	Requirements; UC-01; sequence; activity; domain/state models
Obj. 3 - find produce and report	FR-04/12/13	QS-02/04/09	Requirements; domain model; quality scenarios

Appendix C - Architecture decision record summary

ADR-01 through ADR-04 are summarised in Section 8.3. The authoritative decision basis is the supplied Phase 1 draft and the requirement/use-case artefacts. Editable ADR source files were not present in the Lab 5 archive.

Appendix D - Submission evidence register

Artefact	Included in Lab 5 package?	Use in report
README.md	Yes	Project overview, repository identity, core scope.
Requirements.md	Yes	Functional requirements and traceability.
acceptance-criteria.md	Yes	AC-01 to AC-05.
quality-scenarios.md	Yes	QS-01 to QS-10.
Glossary.md	Yes	Terminology.
UC.md	Yes	Detailed UC-01.
domain-model.png	Yes	Domain analysis.
Sequence diagram.drawio.png	Yes	Interaction model.
activity.drawio.png	Yes	Workflow model.
state machine diagram.png	Yes	Lifecycle model.



Artefact	Included in Lab 5 package?	Use in report
Phase 1 draft.pdf	Yes	Integrated Phase 1 baseline.
Application source code	No	Implementation evidence pending.
Executed automated test logs	No	Runtime evidence pending.
Completed UI wireframes/usability recordings	No	Usability evidence pending.